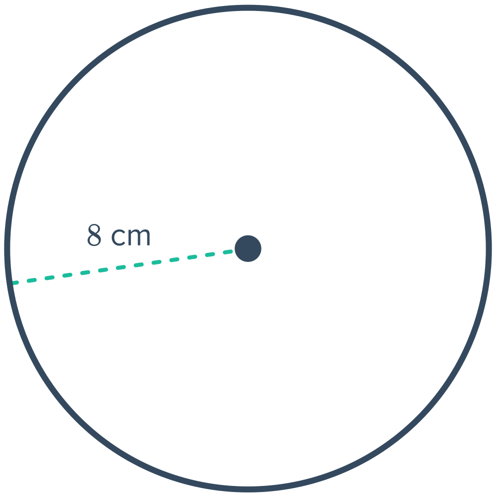
Area of a circle

1 For each of the following circles:

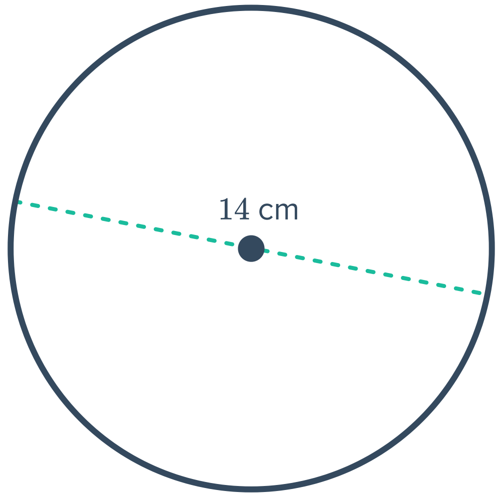
i Calculate the exact area.

ii Hence, calculate the area rounded to two decimal places.

a



b



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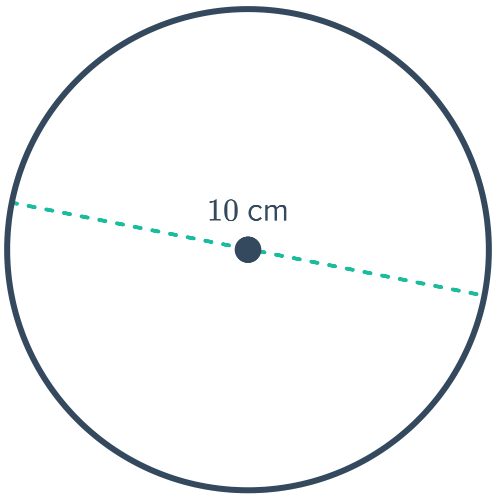

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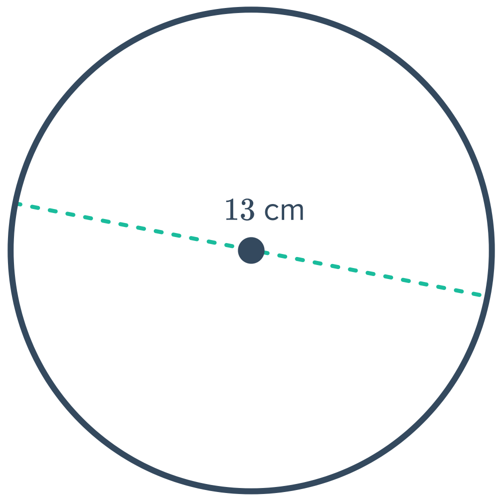

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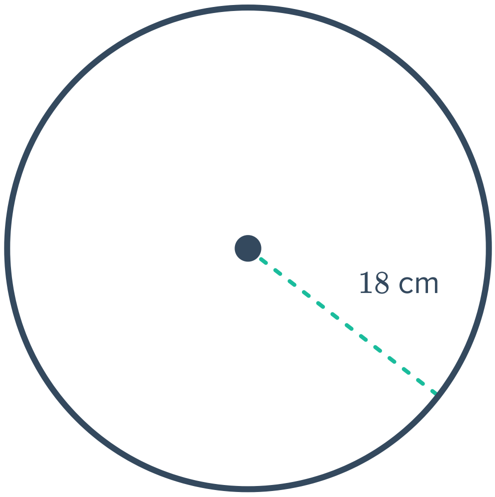

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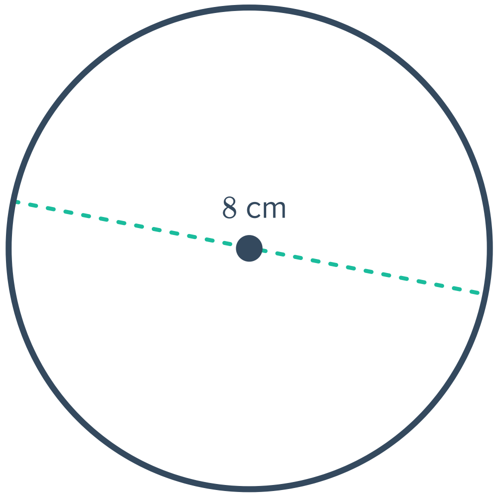


2 Calculate the exact area of the following circles:

a A circle has a radius of $4\,\mathrm{mm}$.

b A circle has a diameter of $8\,\mathrm{mm}$.

c A circle has a radius of $12\,\mathrm{mm}$.

3 Calculate the area of the following circles, rounded to two decimal places:

a A circle has a radius of $9\ \mathrm{cm}$.

b A circle has a diameter of $24\ \mathrm{cm}$.

c A circle has a radius of $12\,\mathrm{mm}$.

4 For each of the following circles:

i Calculate the radius.

ii Hence, calculate the exact area.

a A circle has a diameter of $12\,\mathrm{mm}$.

b A circle has a circumference of $14\pi\,\mathrm{cm}$.

c A circle has a circumference of $18\,\mathrm{cm}$.

d A circle has a diameter of $22\,\mathrm{mm}$.

5 Find the exact radius of the following circles:

a A circle with exact area of $64\pi\ \mathrm{cm}^2$.

b A circle with area of $36\ \mathrm{cm}^2$.

c A circle with area of $36\pi\ \mathrm{cm}^2$.

d A circle with area of $16\pi\ \mathrm{cm}^2$.

e A circle with area of $25\,\mathrm{mm}^2$.

6 For each of the following circles:

i Calculate the radius.

ii Hence, calculate the exact circumference.

a A circle with area of 25

b A circle has an area of 121

7 The area of a circle is 352 cm^2.

a If its radius is $r \text{ cm}$, find r, correct to two decimal places.

b Hence, find the circumference of the circle, correct to one decimal place.

8 A circle has an area of 144 mm^2.

a Calculate the exact radius.

b Hence, calculate the radius rounded to two decimal place.

9 A circle has an area of $16\pi \text{ cm}^2$.

a Calculate the radius.

b Hence, calculate the diameter.

10 A circle has an area of 25 mm^2.

a Calculate the exact radius.

b Calculate the exact diameter.

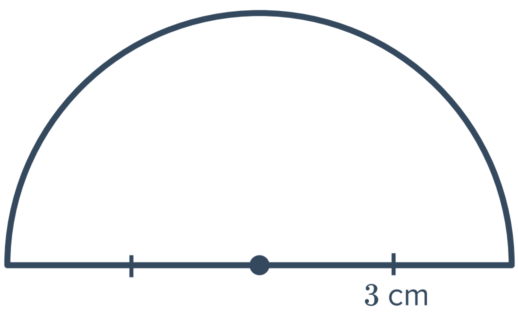
11 A cicle has an area of 144 mm^2.

a Calculate the exact diameter.

b Hence, calculate the diameter rounded to two decimal places.

12 Find the area of the following shapes to one decimal place:

a



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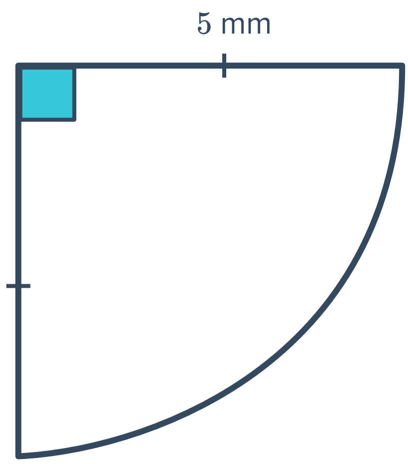

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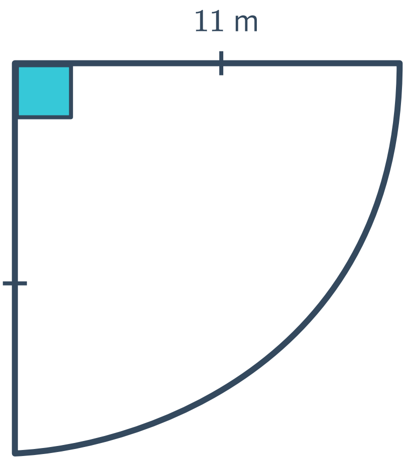

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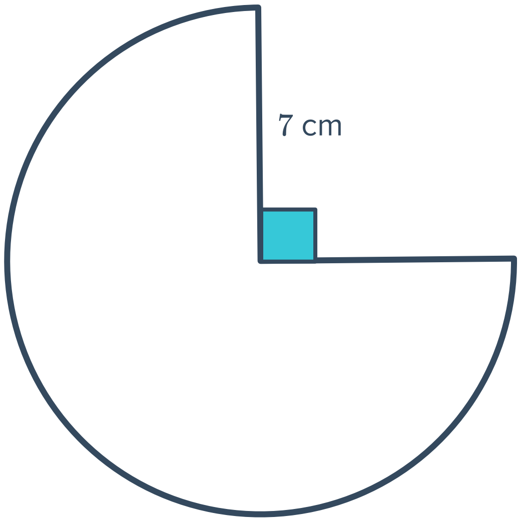
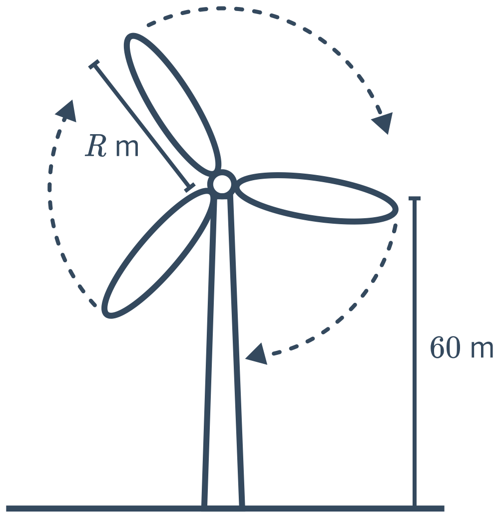


Applications

13 The radius of a circular baking tray is 10 cm. Find its area, correct to two decimal places.

14 A wind turbine has blades that are $R \text{ m}$ long which are attached to a tower 60 m high. When a blade is at its lowest point (pointing straight down), the distance between the tip of the blade and the ground is 20 m.



a Calculate the value of R.

b Find the distance travelled by the tip of the blade during one full revolution, correct to two decimal places.

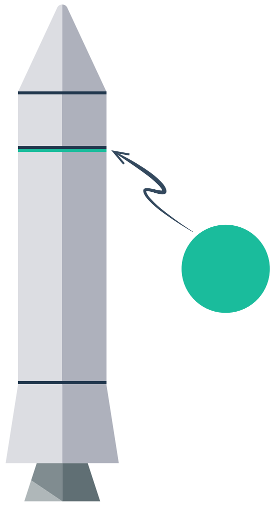
c A factor in the design of wind turbines is the amount of area covered by their blades. The larger the area covered, the more air can pass through the blades.

Find the area inside the circle defined by the rotation of the blade tips, correct to two decimal places.

15 The engineering team at Rocket Surgery are building a rocket for an upcoming Mars mission.

A critical piece is the circular connective disc that connects the booster rocket to the rest of the spacecraft. This disc must completely cover the top of the booster rocket.

The booster rocket has a diameter of precisely 713.5 centimetres. Answer the following correct to two decimal places.



a Find the required area of the connective disc.

b Instead of using the exact value, an engineer uses the approximation 3.14 for π.

Find the area using the engineer's approximation for π.

c If the connective disc is more than $100\ \mathrm{cm}^2$ too small, the disc will malfunction, resulting in catastrophic launch failure.

Will the disk malfunction if it is built according to the engineer's calculation? Explain your answer.